Algebra 1

2024 / 2025

Duration : 3h

FINAL EXAM

Exercise 01 (4 marks)

Let α and β be real numbers, and let $P(X) = X^6 + 4X^5 + 8X^4 + 10X^3 + \alpha X^2 + \beta X + 1 \in \mathbb{R}[X]$ which admits -1 as a double root.

1. Determine α and β.
2. Let j be the root of $X^3 - 1$ with a positive imaginary part. Show that j is a double root of P.
3. Factorize P into irreducible factors, first in $\mathbb{C}[X]$, then in $\mathbb{R}[X]$.
4. Give the Taylor expansion of P at 1.
5. Perform the division by increasing power order of order 2 of $P(X)$ by $Q(X) = X^2 + 1$ in $\mathbb{R}[X]$.

Exercise 02 (16 marks)

In this exercise, a "ring" refers to a ring with unity. Let $(R, +, \cdot)$ and $(S, +, \cdot)$ be two non-trivial rings. We respectively denote by $U(R)$, $D(R)$, and $I(R)$ the set of invertible elements of R, the set of zero divisors of R, and the set of idempotent elements of R.

I. In this part, we define addition and multiplication on the Cartesian product $R \times S = \{(r, s) : r \in R, s \in S\}$ as follows : $(r_1, s_1) + (r_2, s_2) = (r_1 + r_2, s_1 + s_2)$ and $(r_1, s_1) \cdot (r_2, s_2) = (r_1 \cdot r_2, s_1 \cdot s_2)$. It is straightforward to verify that $R \times S$ is a ring under these binary operations (called the **Canonical** ring structure on $R \times S$).

(a) 1. Show that $R \times S$ is commutative if and only if both R and S are commutative.

2. Show that $R \times S$ is not an integral domain.
3. Show that $U(R \times S) = U(R) \times U(S)$.
4. Show that $I(R \times S) = I(R) \times I(S)$.
5. Determine $D(R \times S)$.
6. Let J be an ideal of $R \times S$. Define : $J_1 = \{x \in R : (x, 0) \in J\}$ and $J_2 = \{y \in S : (0, y) \in J\}$.
   - i. Show that J_1 is an ideal of R and J_2 is an ideal of S.
   - ii. Prove that $J = J_1 \times J_2$.
   - iii. Deduce that every ideal of $R \times S$ is of the form $I \times J$, where I and J are ideals of R and S, respectively.

(b) We now take $R = S = \mathbb{Z}$.

1. Determine $U(\mathbb{Z} \times \mathbb{Z})$, $D(\mathbb{Z} \times \mathbb{Z})$, and $I(\mathbb{Z} \times \mathbb{Z})$.
2. Determine the ideals of $\mathbb{Z} \times \mathbb{Z}$.

3. Identify which of the following sets are ideals and which are subrings of $\mathbb{Z} \times \mathbb{Z}$, justifying your answer : $I_1 = \{(n,n) : n \in \mathbb{Z}\}$, $I_2 = \{(k,\ell) : k,\ell \in \mathbb{Z} \text{ and } \ell - k \text{ is even}\}$, $I_3 = \{(5k, 7\ell) : k,\ell \in \mathbb{Z}\}$.

4. Find all ring homomorphisms from $\mathbb{Z} \times \mathbb{Z}$ to $\mathbb{Z}$.

(c) We take $R = \mathbb{Z}/2\mathbb{Z} = \{\overline{0}, \overline{1}\}$ the ring of integers modulo 2 and equip $R^n = R \times R \times \cdots \times R$ ($n \in \mathbb{N}$) with the binary operations defined as follows : $(\overline{r_1}, \overline{r_2}, \ldots, \overline{r_n}) + (\overline{s_1}, \overline{s_2}, \ldots, \overline{s_n}) = (\overline{r_1 + s_1}, \overline{r_2 + s_2}, \ldots, \overline{r_n + s_n})$ and $(\overline{r_1}, \overline{r_2}, \ldots, \overline{r_n}) \cdot (\overline{s_1}, \overline{s_2}, \ldots, \overline{s_n}) = (\overline{r_1 \cdot s_1}, \overline{r_2 \cdot s_2}, \ldots, \overline{r_n \cdot s_n})$. It is straightforward to verify that $(R^n, +, \cdot)$ is a commutative ring.

1. What is the characteristic of R^n ?
2. Show that $I(R^n) = R^n$.
3. Determine $U(R^n)$ and $D(R^n)$.
4. What are the ideals of R^n ?

II. In this part, we take $R = \mathbb{Z}$ and equip the Cartesian product $\mathbb{Z} \times S$ with the following binary operations :

$$(m,a) \oplus (n,b) = (m+n, a+b) \quad \text{and} \quad (m,a) \otimes (n,b) = (mn, na + mb + ab).$$

(a) 1. Show that $(\mathbb{Z} \times S, \oplus, \otimes)$ is a ring (this ring is called the **Dorroh $\mathbb{Z}$-ring**).

2. What is the neutral element of $\mathbb{Z} \times S$ with respect to $\otimes$?

3. Under what condition is the ring $(\mathbb{Z} \times S, \oplus, \otimes)$ commutative?

4. Let f be the map defined by

$$f : \begin{array}{ccc} S & \longrightarrow & \mathbb{Z} \times S \\ x & \longmapsto & (0, x). \end{array}$$

Is f a ring monomorphism?

5. Let e denote the neutral element of S.

i. Show that $(0, e)$ is the neutral element of $\{0\} \times S$ and a zero divisor in $\mathbb{Z} \times S$.

ii. Let g be the map defined by

$$g : \begin{array}{ccc} \mathbb{Z} \times S & \longrightarrow & S \\ (n, x) & \longmapsto & x + ne. \end{array}$$

Show that g a ring epimorphism and determine $\ker g$.

(b) We take $S = \mathbb{Z}$.

1. Determine the set of invertible elements of $(\mathbb{Z} \times \mathbb{Z}, \oplus, \otimes)$.

2. Determine the set of zero divisors of $(\mathbb{Z} \times \mathbb{Z}, \oplus, \otimes)$.

3. Determine the set of idempotent elements of $(\mathbb{Z} \times \mathbb{Z}, \oplus, \otimes)$.

4. Identify which of the following sets are ideals of $(\mathbb{Z} \times \mathbb{Z}, \oplus, \otimes)$, justifying your answer : I_1, I_2, and I_3.

Final Exam (Answers)

Exercise 1: 1) $P(x) = x^6 + 4x^5 + 8x^4 + 10x^3 + \alpha x^2 + \beta x + 1 \in \mathbb{R}[x]$.

-1 is a double root of $P(x) \Leftrightarrow \begin{cases} P(-1) = 0 \\ P'(-1) = 0 \end{cases} \Leftrightarrow \begin{cases} \alpha - \beta = 4 \\ \beta - 2\alpha = -12 \end{cases}$

$\Leftrightarrow \begin{cases} \alpha = 8 \\ \beta = 4. \end{cases}$ Then $P(x) = x^6 + 4x^5 + 8x^4 + 10x^3 + 8x^2 + 4x + 1$.

2) $j = e^{\frac{2i\pi}{3}}$, j satisfies $j^3 = 1$ and $j^2 + j + 1 = 0$.

Then $P(j) = 12(j^2 + j + 1) = 0$, $P'(j) = 36(j^2 + j + 1) = 0$.

(knowing that $j^{3k} = 1$, $j^{3k+1} = j$, $j^{3k+2} = j^2$, $k \in \mathbb{N}$)

Conclusion: j is a double root of P.

3) since $P(x) \in \mathbb{R}[x]$, $\bar{j} = j^2$ is also a double root of P.

Then:

$P(x) = (x+1)^2 (x-j)^2 (x-j^2)^2$ in $\mathbb{C}[x]$.

$P(x) = (x+1)^2 \left((x-j)(x-j^2)\right)^2 = (x+1)^2 (x^2+x+1)^2$ in $\mathbb{R}[x]$.

4) Taylor expansion of P at 1:

$$P(x) = \sum_{k=0}^{6} \frac{P^{(k)}(1)}{k!} (x-1)^k = 36 + \frac{104}{1!}(x-1) + \frac{274}{2!}(x-1)^2 + \frac{612}{3!}(x-1)^3 + \frac{1032}{4!}(x-1)^4 + \frac{1200}{5!}(x-1)^5 + \frac{720}{6!}(x-1)^6$$

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5) Division by increasing power to order 2

$$P(x) = q(x)\left(1+4x+7x^2\right) + x^3\left(6+x+4x^2+x^2\right).$$

Exercise 2

1) $R \times S$ is Commutative $\Leftrightarrow \forall (r_1, s_1), (r_2, s_2) \in R \times S$

$$(r_1, s_1)(r_2, s_2) = (r_2, s_2)(r_1, s_1) \Leftrightarrow \begin{cases} r_1 r_2 = r_2 r_1 \\ s_1 s_2 = s_2 s_1 \end{cases} \Leftrightarrow$$

R and S are Commutative.

2) Since R and S are no trivial rings, if we take $(r, 0)$ with $r \neq 0$ and $(0, s)$ with $s \neq 0$; we have $(r,0)(0,s) = (0,0)$ even $(r,0) \neq (0,0)$ and $(0,s) \neq (0,0)$.

Then $R \times S$ is not integral (it contains zero divisors).

3/

3) $U(R\times S) \overset{(1)}{=} U(R)\times U(S)$? . Let $(x,y)\in R\times S$.

$(x,y)\in U(R\times S) \Leftrightarrow \exists (x',y')\in R\times S$ such that

$(x,y)(x',y') = (x',y')(x,y) = (e_R; e_S) \Leftrightarrow$

$\begin{cases} x x' = e_R \\ y y' = e_S \end{cases} \Leftrightarrow \begin{cases} x\in U(R) \\ \wedge \\ y\in U(S) \end{cases} \Leftrightarrow (x,y)\in U(R)\times U(S)$. Then (1) is satisfied.

4) $I(R\times S) \overset{(2)}{=} I(R)\times I(S)$? Let $(x,y)\in R\times S$.

$(x,y)\in I(R\times S) \Leftrightarrow (x,y)^2 = (x,y) \Leftrightarrow \begin{cases} x^2 = x \\ y^2 = y \end{cases} \Leftrightarrow$

$\begin{cases} x\in I(R) \\ y\in I(S) \end{cases} \Leftrightarrow (x,y)\in I(R)\times I(S)$, then (2) is

satisfied.

5) $D(R\times S) = ?$. Let $(x,y)\in R\times S$.

$(x,y)\in D(R\times S) \Leftrightarrow \begin{cases} (x,y)\neq(0,0) \\ \wedge \\ \exists (x',y')\neq(0,0) \end{cases}$ such that

$(x,y)(x',y') = (xx', yy') = (0,0)$

$\Leftrightarrow \begin{cases} xx' = 0 \\ yy' = 0 \end{cases} \wedge (x,y)\neq(0,0) \wedge (x',y')\neq 0$.

4/ $D(R\times S) = (D(R)\times S)\cup(R\times D(S))$

$\cup\,((R\setminus\{0\})\times\{0\})\cup(\{0\}\times S\setminus\{0\})$.

6) Let $\bar{J}$ be an ideal of $R\times S$. Let $\bar{J}_1=\{x\in R : (x,0)\in\bar{J}\}$

$0\in\bar{J}_1$ since $(0,0)\in\bar{J}$, then $\bar{J}_1\neq\phi$.

$\begin{cases} x\in\bar{J}_1 \Leftrightarrow (x,0)\in\bar{J} \\ x'\in\bar{J}_1 \Leftrightarrow (x',0)\in\bar{J}\end{cases}$. since $\bar{J}$ is an ideal $\underbrace{(x,0)-(x',0)}_{(x-x',0)}\in\bar{J}$

hence $x-x'\in\bar{J}_1$;

Let $x\in\bar{J}_1$ and $\alpha\in R$. $x\in\bar{J}_1\Leftrightarrow(x,0)\in\bar{J}$, $\alpha\in R \Leftrightarrow (\alpha,0)\in R\times S$.

since $\bar{J}$ is an ideal $(\alpha,0)(x,0)=(\alpha x,0)\in\bar{J}$; then $\alpha x\in\bar{J}_1$.

We conclude that $\bar{J}_1$ is an ideal of R.

Using the same technique we prove that $\bar{J}_2$ is an ideal of S.

Let $J_1 \times J_2 \overset{?}{=} J$. We prove double inclusion.

$(x,y) \in J_1 \times J_2$, then

$$\begin{cases} x \in J_1 \\ \wedge \\ y \in J_2 \end{cases} \Leftrightarrow \begin{cases} (x,0) \in J \\ \wedge \\ (0,y) \in J \end{cases} \Rightarrow (x,0)+(0,y)=(x,y) \in J.$$

(Since J is an ideal)

Conversely, $(x,y) \in J \Rightarrow \forall (\alpha,\beta) \in R \times S$

$(\alpha,\beta)(x,y) \in J$, We take $(\alpha,\beta)=(1,0)$ then

$(1,0)(x,y) = (x,0) \in J$.

Hence $(x,0) \in J$, then $x \in J_1$.

We take $(\alpha,\beta)=(0,1)$; we obtain $(0,y) \in J$; then

$y \in J_2$. Conclusion $(x,y) \in J_1 \times J_2$.

Conclusion: $J = J_1 \times J_2$.

Then; any ideal of $R \times S$ is of the form

$J_1 \times J_2$, where J_1 ideal of R, J_2 ideal of S.

b/

Applying part a)

b) 1) $U(\mathbb{Z}\times\mathbb{Z}) = U(\mathbb{Z})\times U(\mathbb{Z})$ (following a)-3)

$= \{-1,1\}\times\{-1,1\}$ $(U(\mathbb{Z}) = \{1,-1\})$

$= \{(1,1),(1,-1),(-1,1),(-1,-1)\}$

$I(\mathbb{Z}\times\mathbb{Z}) = I(\mathbb{Z})\times I(\mathbb{Z}) = \{0,1\}\times\{0,1\}$ $(I(\mathbb{Z}) = \{0,1\})$

$= \{(0,0),(0,1),(1,0),(1,1)\}$

$D(\mathbb{Z}\times\mathbb{Z}) = \mathbb{Z}\setminus\{0\}\times\{0\} \cup \{0\}\times\mathbb{Z}\setminus\{0\}$. ($\mathbb{Z}$ is integral)

2) Following 6.iii) the ideals of $\mathbb{Z}\times\mathbb{Z}$ are

$n\mathbb{Z}\times m\mathbb{Z}$, $n,m\in\mathbb{N}_0$.

3) I_1 is a subring; but it is not an ideal since $(2,3)\in\mathbb{Z}\times\mathbb{Z}$, $(1,1)\in I$ but $(2,3)(1,1)=(2,3)\notin I_1$.

I_2 is not an ideal since $(3,5)\in I_2$, $(3,2)\in\mathbb{Z}\times\mathbb{Z}$ but $(3,5)(3,2)=(9,10)\notin I_2$. I_2 is a subring (easy).

I_3 is not a subring $((1,1)\notin I_3)$. I_3 is an ideal.

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7) $f : \mathbb{Z}\times\mathbb{Z} \longrightarrow \mathbb{Z}$ is homomorphism of rings.

$f(n,m) = nf(1,0) + mf(0,1)$; $\forall n,m \in \mathbb{Z}$

and $f(1,1) = 1$. Then $f(1,0) = 1$

$f(0,1) = 0$

or $f(1,0) = 0$

$f(0,1) = 1$.

Which gives $f_1 : \mathbb{Z}\times\mathbb{Z} \longrightarrow \mathbb{Z}$

$(n,m) \longrightarrow n$

or $f_2 \; (n,m) \longrightarrow m$.

Conversely, it easy to verify that f_1 and f_2 are ring homomorphisms

c) $\forall (\bar{x}_1, \dots, \bar{x}_n) \in R^n$, we have

$(\bar{x}_1, \dots \bar{x}_n) + (\bar{x}_1, \dots \bar{x}_n) =$

$(2\bar{x}_1, \dots 2\bar{x}_n) = (\bar{0}, \bar{0}, \bar{0} \dots \bar{0})$ (Since caracteristic of R is 2)

then the caracteristic of R^n is 2.

8)

Applying the part a) we get easily:

$I(R^n) = (I(R))^n$ (Cartesian product) since $\overline{0}^2 = \overline{0}$ and $\overline{1}^2 = \overline{1}$

$= R^n$

$U(R^n) = (U(R))^n$ (Cartesian product)

$= (\{\overline{1}\})^n = \{(\overline{1}, \overline{1}, \ldots \overline{1})\}$.

$D(R^n) = R^n \setminus \{(\overline{1}, \overline{1}, \ldots \overline{1})\}$.

The ideals of R^n are of the form

$\prod_{i=1}^{n} S_i \quad S_i \in \{\{\overline{0}\}, R\}$ (since the only ideals of R are $\{\overline{0}\}$ and R.)

II) $(\mathbb{Z} \times S, \oplus, \otimes)$ is a ring (simple calculation)

$(\mathbb{Z} \times S, \oplus, \otimes)$ is a Commutative ring ($\Rightarrow (S, +, \cdot)$ is a commutative ring.

The neutral element is $(1, 0)$.

$f(e) \neq (1,0)$, then f is not monomorphism.

$\forall x \in S$, $(0, e) \otimes (0, x) = (0, e \cdot x) = (0, x)$, then $(0, e)$ is neutral element in $\{0\} \times S$.

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9)

$(0,e) \otimes (m,a) = (0, me + ea)$.

then $(0,e) \otimes (m, -me) = (0, me - me) = (0,0)$.

then $(0,e)$ is a zero divisor $\left(\begin{matrix} m \in \mathbb{Z} \\ m \neq 0 \end{matrix}\right)$

in $\mathbb{Z} \times S$.

$$\underbrace{g\left((n,x) \oplus (m,y)\right)}_{x+y+(n+m)e} = g(n,x) \oplus g(m,y)$$

$$= \underbrace{x + ne + y + me}_{x+y+(n+m)e}$$

$$g\left((n,x) \otimes (m,y)\right) = g(nm, mx + ny + xy)$$

$$= mx + ny + xy + nme.$$

$$g(n,x) \cdot g(m,y) = (x+ne)(y+me).$$

$$= xy + mx + ny + nme.$$

$\forall\ x \in S$; $\exists (\alpha, \beta) \in \mathbb{Z} \times S$, $\alpha = 0$, $\beta = x$

with $g(\alpha, \beta) = x$; g is surjective.

Conclusion g is epimorphisme of rings.

$$\ker g = \{(n, -ne);\ n \in \mathbb{Z}\}$$

10/ b)

$(m,a) \in \mathbb{Z}\times\mathbb{Z}$ is invertible in $(\mathbb{Z}\times\mathbb{Z}, \oplus, \otimes)$ iff

$\exists (n,b) \in \mathbb{Z}\times\mathbb{Z}$ such that $(m,a)\otimes(n,b) = (1,0)$

$$\iff \begin{cases} mn = 1 \\ na + mb + ab = 0 \end{cases} \iff \begin{cases} m = 1 \wedge n = 1 \\ a + b + ab = 0 \end{cases} \text{ or}$$

$$\begin{cases} m = n = -1 \\ -a - b + ab = 0. \end{cases}$$

This gives the set of invertible elements:

$$\{(1,0), (1,-2), (-1,0), (-1,2)\}$$

The set of idempotent elements:

$(m,a) \in \mathbb{Z}\times\mathbb{Z}$ is idempotent $\iff (m,a)\otimes(m,a) = (m,a)$

$$\iff \begin{cases} m^2 = m \\ 2ma + a^2 = a \end{cases} \iff \begin{cases} m = 1 \vee m = 0 \\ 2ma + a^2 = a \end{cases} \iff$$

$$\begin{cases} m = 1 \\ 2a + a^2 = a \end{cases} \vee \begin{cases} m = 0 \\ a^2 = a \end{cases} \text{ ie } \begin{cases} m = 1 \\ a(a+1) = 0 \end{cases} \vee \begin{cases} m = 0 \\ a^2 = a \end{cases}$$

$$\{(0,0), (0,1), (1,0), (1,-1)\}.$$

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The zero divisors : (m,a) is zero divisor if $(m,a) \neq (0,0)$ such $\exists (n,b) \neq (0,0)$ that

$(m,a) \otimes (n,b) = (0,0) \iff$

$$\begin{cases} mn = 0 \\ na + mb + ab = 0 \end{cases} \iff \begin{cases} m = 0 \wedge a \neq 0 \\ na + ab = 0 \end{cases} \vee$$

$$\begin{cases} n = 0 \wedge b \neq 0 \\ mb + ab = 0 \end{cases}$$

ie

$$\begin{cases} m = 0 \wedge a \neq 0 \\ a(n+b) = 0 \end{cases} \vee \begin{cases} n = 0 \wedge b \neq 0 \\ b(m+a) = 0 \end{cases}$$

$$\{(0,k)\ k \neq 0\} \cup \{(k,-k)\ k \in \mathbb{Z}\ k \neq 0\}.$$

I_1 is not an ideal, I_2 is an ideal, I_3 is not an ideal (simple verifications)

all of I_1, I_2, I_3 are subgps of $(\mathbb{Z}\times\mathbb{Z}, \oplus)$, for I_1 and I_3 the condition $(\alpha,\beta) \in \mathbb{Z}\times\mathbb{Z}, (m,a) \in \mathbb{Z}\times\mathbb{Z}$

$\Longrightarrow (\alpha,\beta) \otimes (m,a) \in I_3$ or I_1 is not satisfied.